

HANOI PEDAGOGICAL UNIVERSITY 2

FACULTY OF ENGLISH



Final Examination: History of World Civilization

Student: Lê Cẩm Giang

Topic: Exam code 07

Class: HIS201.01

Student's code: 257140231038

PHU THO, 2026

CONTENTS

CONTENTS	i
I. TASK 1	1
II. TASK 2	3
1. Scholastic philosophy and the emergence of the university system	3
1.1. Scholastic philosophy	3
1.2. The emergence of the university	4
2. The evolution of structure and aesthetics in medieval architecture	5
2.1. Romanesque architecture	5
2.2. Gothic architecture	6
3. Art	6
3.1. Illuminated manuscript	6
3.2. The transformation of painting through Giotto	7
4. Conclusion	8
III. TASK 3	8
1. Transfer of thinking systems: From Theocentrism to Anthropocentrism	8
1.1. The medieval worldview and the limits of scholastic philosophy	8
1.2. The rise of Renaissance humanism	9
1.3. The consequence of the formation of modern thought	9
2. Astronomical revolution and collapse of the classical universe model	9
2.1. The Ptolemaic Geocentric model and religious authority	10
2.2. Nicolaus Copernicus and the heliocentric hypothesis	10
2.3. The experimental proof of Galileo Galilei	11
3. The revolution in anatomy and the mechanization of the human body	12
3.1. The Galenic humoral theory in medieval medicine	12
3.2. The breakthrough of Andreas Vesalius	12
3.3. Leonardo da Vinci: The intersection of art and mechanical inquiry ..	12
4. The information revolution and the role of Gutenberg's printing press	13
4.1. The information crisis of manuscript culture	13

4.2. The emergence of the Republic of Letters	14
5. Conclusion	14
REFERENCE	16

I. TASK 1 (2 points): Write a short paragraph (about 500 words) about the pyramids in Ancient Egypt and explain their significance to Egyptian civilization.

The pyramids in ancient Egypt are typical architectural works, associated with the beliefs and social organization of this civilization. Built mainly during the Old Kingdom (about 2686-2181 BC), this enormous tomb complex is situated on the west side of the Nile, which was regarded by Ancient Egyptians as the boundary between the world of the living and the land of the dead. In terms of history and geography, these structures reflect a great evolution of geometric thinking. Starting in 2630 BC under Pharaoh Djoser (3rd Dynasty) with the Step Pyramid at Saqqara, this technique was perfected around 2575 BC under the reign of Pharaoh Sneferu (4th Dynasty) at Dahshur to create the first perfect pyramid.



Picture 1: The Step Pyramid of Djoser 1

The peak of this era was the Giza Complex which was built between 2560 and 2500 BC as the resting place for three generations of Pharaohs: Khufu, Khafre and Menkaure. The Great Pyramid of Khufu, known as the greatest masterpiece, originally stood 146.6 meters high and held the record as the world's tallest man-made structure for more than 3,800 years. With a base area covering 5.3 hectares,

the project is made up of about 2.3 million blocks of limestone and marble, each block weighing an average of 2.5 to 15 tons, creating a huge total volume estimated at more than 5.7 million tons [1]. This grandeur of size goes hand in hand with archaeological mysteries. The ancient Egyptians positioned the four faces of the pyramid to face the four cardinal directions of the compass with an astonishing error of less than 1/12 of a degree [2]. Inside the structure is a complex system of tunnels and rooms, including the Chamber of God and the Great Hall, which are purposely aligned with constellations such as Orion and Sirius.



Picture 2: The Giza Pyramid Complex.

Until now, the method of transporting and lifting stone blocks weighing tens of tons from quarries hundreds of kilometers away in Tura and Aswan to a height of hundreds of meters is still a mystery. Although there are many hypotheses about the system of straight slopes or internal spiral ramps, archaeologists have not yet found any absolute material evidence from the Old Kingdom to confirm their specific technique. However, the real significance of the pyramids lies not in the blocks of stone, but in their core role in Egyptian civilization. In terms of religion, Egyptologist Mark Lehner (1997) argued that the pyramid acted as a "rebirth machine" for the Pharaohs[3]. Ensuring the immortality of the king is a precondition for maintaining Ma'at (order, justice and cosmic balance), helping the country avoid

natural disasters and chaos. From a political and economic perspective, pyramid construction played a crucial role in strengthening the administrative system of Ancient Egypt. Managing a construction project that lasted for decades and involved 20,000 to 30,000 workers required the Pharaoh's administration to develop efficient systems of taxation, resource management, and food distribution across the kingdom. Archaeological excavations at Heit el-Ghurab in Giza have challenged the long-standing belief that the pyramids were built by oppressed slaves. Instead, the evidence suggests that the workforce consisted largely of skilled craftsmen and laborers who were housed, fed, and supported by the royal administration (Hawass, 2006; Murray, 2004) [4][5]. This process of conscription transformed a society of discrete communities along the Nile into a highly centralized state, creating the first bureaucratic systems in human history (Kemp, 2006) [6]. Therefore, the pyramid is not only an architectural work but also a symbol of religion, political power and the level of development of Ancient Egypt.

II. TASK 2 (3 points): Write a short essay (about 1.000 words) analyzing the major achievements of Medieval European culture, such as art, architecture, or philosophy. Use relevant examples to support your ideas.

The Middle Ages (around the 5th to the 15th century) was long described by later historians as the "*Dark Ages*", a period of decline and stagnation in Europe after the Roman Empire collapsed. However, modern historians have proven that this is a strong transformation era, which laid a solid foundation for Western civilization. Under the assistance and ethics of the Catholic Church, Medieval European Culture achieved brilliant and groundbreaking achievements, especially in the fields of scholastic philosophy, church architecture and religious art.

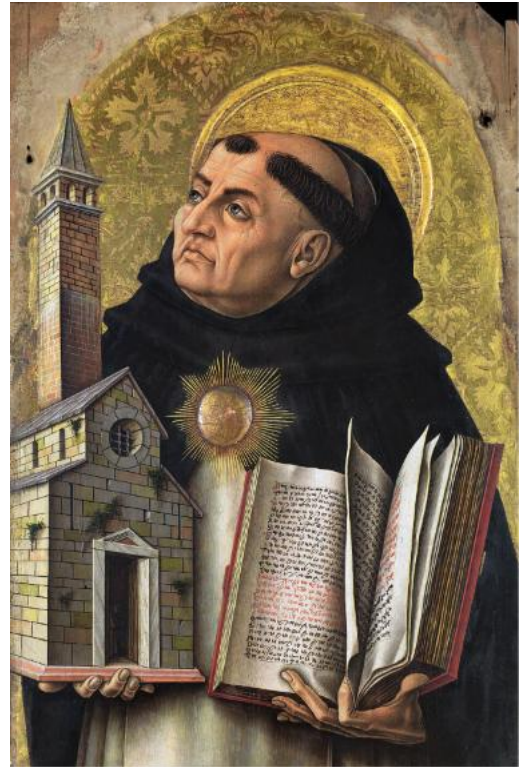
1. Scholastic philosophy and the emergence of the university system

1.1. Scholastic philosophy

¹ The Dark Ages refers to the Early Middle Ages in Europe (c.5th-10th centuries), a period once thought to be marked by cultural and intellectual decline following the fall of the Western Roman Empire.

One of the greatest intellectual achievements of the Middle Ages was the formation of scholasticism. Unlike the misconception that this period restricted thinking, medieval scholars implemented a major intellectual mission: the harmony between Christian faith and the ancient Greek philosophy, especially the thought of *Aristotle*.²

A prominent representative of this movement was Thomas Aquinas (1225-1274) with his classic work: *Summa Theologiae* (*Summary of Theology*). Aquinas argued that reason and faith were not in conflict but worked together [7]. Human reason could help people understand the laws of nature, thereby leading them closer to God's truth. The rigorous academic argument method (dialectic) of scholastic philosophy stimulated critical thinking, becoming a premise for modern Western philosophy.



Picture 3: Thomas Aquinas

1.2. The emergence of the university

The rise of scholastic philosophy went hand in hand with the introduction of a new educational institution: the university system. The earliest universities in the world such as Bologna University (Italy, 1088), Paris University (France, 1150) or Oxford University (UK, 1167) were established in this period.

The universities functioned as the autonomous communities of teachers and students. They were academic centers where students from different parts of

² Aristotle was an ancient Greek philosopher and polymath (384-322 BCE)

Europe studied law, medicine, theology and the “*Seven Liberal Arts*”³. This medieval university system was the cradle of intellectuals, preparing the educational and philosophical foundations for the explosion of the Renaissance.

2. The evolution of structure and aesthetics in medieval architecture

2.1. Romanesque architecture

Architecture reflects clearly worldwide and technical capacity of the Middle Ages. Developing in the early stages of the medieval period (XI-XII centuries), the Romanesque style was greatly influenced by ancient Roman architecture, reflecting Europe's need for protected sacred spaces.

Romanesque churches, such as Saint-Sernin church in Toulouse (France), were characterized by semicircular arches, solid stone walls and massive pillars. Because construction techniques depended entirely on the main bearing walls, the windows of Romanesque architecture were relatively small and narrow. The spaces inside were often dark, evoking a strict, static and closed atmosphere, resembling a strong spirit fortress to protect people before the mundane forces.



³ The Seven Liberal Arts: The standard curriculum of medieval education, consisting of the *trivium* (grammar, rhetoric, and logic) and the *quadrivium* (arithmetic, geometry, music, and astronomy).

Picture 4: Saint-Sernin Basilica, a masterpiece of Romanesque architecture.

2.2. Gothic architecture

By the mid-12th century, a revolution in construction techniques gave birth to the Gothic style, originating from the reconstruction of the Basilica of Saint-Denis in France. Medieval architects invented three breakthrough technical solutions: the ribbed vault, the flying buttress and the pointed arch. By moving the entire gravity of the dome out of independent pillars outside the building, Gothic architects no longer needed massive load-bearing walls. As a result, the walls were released, turning into giant stained glass windows (Stained Glass) with maximum natural light.



Churches like Chartres Cathedral or Notre-Dame De Paris rose to the sky with slender, sharp towers. Gothic spaces were filled with vibrant colors from glass paintings, transforming natural light into *Lux Nova*⁴ ("New Light"), elevating the souls of visitors toward God and heaven.

3. Art

3.1. Illuminated manuscript

Medieval art primarily served religious and educational purposes for the masses, especially before the rise of widespread literacy, given that much of the population at the time was illiterate. Therefore, this artistic period did not prioritize the precise reproduction of the natural world but focused on symbolism.

⁴ *Lux Nova* ("new light"): A Gothic architectural concept in which light, especially through stained glass, symbolizes the presence of God and spiritual enlightenment.

Outstanding achievements were first in the art of illuminated manuscripts. The monks in the monasteries devoted their lives to hand-copying and illustrating the Bible using gold and silver leaf and pigments extracted from natural minerals. Books like Book of Kells were invaluable visual art masterpieces, preserving knowledge and literary traditions through centuries of political and social change.



Picture 6: Pages from the Book of Kells.

3.2. The transformation of painting through Giotto

Medieval sculpture and post-art painting began to move toward naturalism. In the 13th century, the sculptures on the domes of Gothic were no longer rigid, flat as in earlier periods and became increasingly lifelike, displaying more realistic facial expressions and flowing drapery.

In paintings, Giotto di Bondone (1267-1337) initiated a profound artistic transformation through his frescoes in the Scrovegni Chapel in Italy. In his works,

he began to put a convincing sense of spatial depth and emotionally intense humanities into religious characters. Giotto's innovations served as a crucial bridge, paving the way for the later peak of Renaissance painting.

4. Conclusion

In conclusion, European culture in the Middle Ages was far from the cultural gap suggested by the outdated prejudice about the “Dark Ages”. Rather, it was a dynamic period of explosive growth in creativity and directed innovation, laying the most solid foundation for Western civilization.

The emergence of scholastic philosophy and the university system shaped the Physical Science Methodology. The magnificent Gothic cathedrals demonstrated a great step on technical thinking and structural elegance. Meanwhile, medieval art successfully completed the role of cultural preservation and infused symbolic forms with profound spiritual meaning. The rich legacy of this period served as the indispensable foundation to support the entire revival and modern era of humanity.

III. TASK 3 (5 points): Write an essay (about 1.500 words) evaluating how the Renaissance shaped modern science and thought. Provide specific examples to support your analysis.

1. Transfer of thinking systems: From *Theocentrism*⁵ to *Anthropocentrism*⁶

1.1. The medieval worldview and the limits of scholastic philosophy

Before the Renaissance, European thought was controlled by Theocentrism. Under this methodology, God was the center of all reality, and all knowledge about nature had to follow the religious rules.

⁵ Theocentrism is the belief that God is the central focus of all existence, guiding purpose, values, and human actions.

⁶ Anthropocentrism is the belief that human beings are the central or most important entity on the planet

Page 8

Humans were considered as passive creatures with “Original Sin”; earthly life was only a challenging stage to prepare for salvation in the afterlife. All intellectual activities during this period primarily aimed to explain and protect for truths of faith available in the Bible.

1.2. The rise of Renaissance humanism

As Nauert (2006) analyzed, scholars in this period rebuilt, edited and critically examined the original ancient documents [8]. They did not religion, but they shifted the center of awareness from God to humanity (Anthropocentrism).

In the *Oration on the Dignity of Man* (1486), philosopher Giovanni Pico Della Mirandola argued that God gave people freely willing and the ability to determine their own [9]. Humans were no longer bound in the feudal and religious society but became people who could make rational decisions and develop their abilities through the study of Studia Humanitatis including: grammar, history, poetry and moral philosophy (Nauert, 2006) [8].

1.3. The consequence of the formation of modern thought

This transformation was crucial for the development of modern thought because of two reasons. Firstly, it is legalizing the study of the secular world. If nature was the creation of a rational God, the fact that people used reason to research nature was no longer against religious beliefs, but rather a method of understanding the objective rules of the universe. The second reason is laying the foundation for modern Individualism. The Renaissance inspired concepts of human rights and freedom of thought, which later became pillars of modern political and social institutions.

2. Astronomical revolution and collapse of the classical universe model

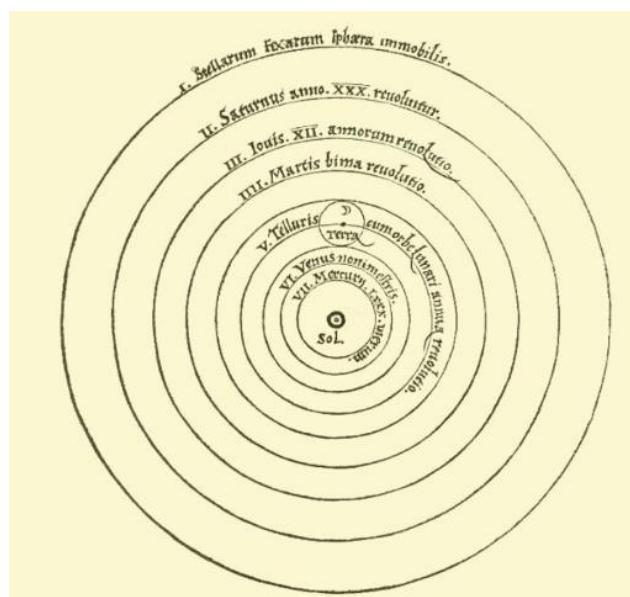
2.1. The Ptolemaic Geocentric model and religious authority

Before the Renaissance, the Geocentric Model of Claudius Ptolemy combined with Aristotle physics, was a standard knowledge system that was sponsored by the Catholic Church. This system assumed the Earth as a fixed center of the universe, surrounded by concentric celestial spheres containing the Moon, the Sun and the planets moving in the perfect circular orbits.

This model confirmed the superior position of the in God's creative plan. According to Kuhn (1957), any shaking of this model also threatens the structure of the powers and the correctness of the principle [10].

2.2. Nicolaus Copernicus and the heliocentric hypothesis

In 1543, the Polish astronomer Nicolaus Copernicus published *De Revolutionibus Orbium Coelestium (On the Revolutions of the Heavenly Spheres)*, introducing the heliocentric model. By applying complex geometric calculations to resolve inconsistencies in planetary predictions within the geocentric system, Copernicus (1543/1992) proposed that the Sun was at the center of the universe, while Earth was considered a planet that rotated on its axis and orbits the Sun [14]. This step basically altered the mathematical structure of astronomy, even though he still had to rely on epicycles to explain these movements.



Picture 7: Copernicus's heliocentric model.

2.3. The experimental proof of Galileo Galilei

Although Copernicus's hypothesis was initially in conflict because people did not feel the Earth moved, the revolution was proven experimentally by Galileo Galilei at the beginning of the 17th century. In *Dialogue Concerning The Two Chief World Systems* (1632/1967), Galileo used an improved telescope to observe the sky [15].



Picture 8: Galileo's telescope

His discovery undermined Aristotelian physics and medieval metaphysics:

- Sunspots and the rugged lunar surface: demonstrated that celestial bodies were not perfect and immutable.
- The four moons of Jupiter: showed that Earth was not the sole center of motion in the universe.
- The phases of Venus: provided direct empirical evidence supporting the Copernican system and refuting the Ptolemaic model.

According to Kuhn (1957), moving the Earth out of the universe forced philosophers to think about humanity in space, establishing a modern principle: Scientific truth must be based on the combination of mathematical models and experimental observations, not on ancient texts [10].

3. The revolution in anatomy and the mechanization of the human body

3.1. The Galenic humoral theory in medieval medicine

Alongside the transformation of cosmology, the Renaissance witnessed a revolution in understanding the human body. During the Middle Ages, medicine was dominated by the theories of Galen, based on the *Humorism*⁷ system (and surgery from animals (pigs and monkeys). Medieval theology considered the body as a sacred structure, contributing to the persistence of misconceptions for over a millennium.

3.2. The breakthrough of Andreas Vesalius

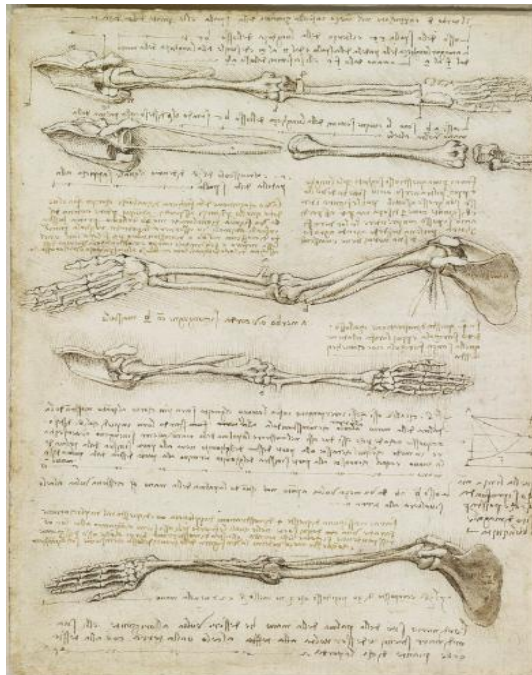
A major breakthrough occurred in 1543 when anatomist Vesalius published *De Humani Corporis Fabrica (On the Fabric of the Human Body)*. Vesalius (1543/2003) changed research methods by directly dissecting the human body to verify theory [11].

In *Fabrica*, Vesalius indicated and modified more than 200 serious errors of Galen on bone structure, blood vessels and nerves. He argued that medical knowledge should be derived not from unquestioned texts but from direct observation and examination.

3.3. Leonardo da Vinci: The intersection of art and mechanical inquiry

The contribution of Renaissance art played an important role in the great. Typical scientists like Leonardo Da Vinci strictly combined aesthetic thinking and experimental surveys. As *Clayton and Philo (2012)* prove, Leonardo Da Vinci conducted more than 30 human dissections, recorded by absolutely precise technical drawings of muscles, bones and circulations [12].

⁷ Humorism was an ancient medical system that posited that the human body was governed by four essential fluids or humors: blood, yellow bile, black bile, and phlegm.



Picture 9: Leonardo da Vinci - The bones and muscles of the arm

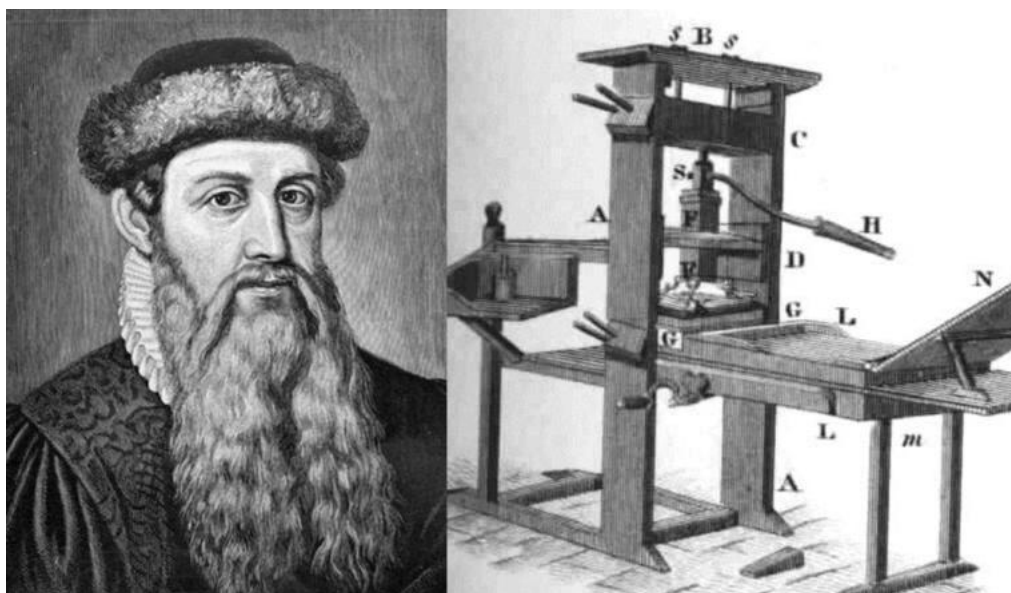
These drawings were the first mechanical studies of human movement, considering joints such as levers and muscles (Clayton & Philo, 2012) [12]. This shift laid the foundation for evidence-based medicine, replacing the concept of disease with the Mechanistic View: The human body is a complex material system, operating under objective physiological and mechanical laws.

4. The information revolution and the role of Gutenberg's printing press

4.1. The information crisis of manuscript culture

A crucial yet often overlooked factor in history is the technical infrastructure that allowed the spread of knowledge. The development of science required a continuous and systematic means of scholarly communication. This system was made possible by the invention of the movable-type printing press by Johannes Gutenberg around 1440 in Mainz (Germany).

Previously, the copying of texts relied entirely on manual handwriting by monks. This process was slow, expensive and caused the number of limited copies. In addition, the rate of human error accumulated over generations, leading to distorted scientific texts (Eisenstein, 1980) [13].



Picture 10: Gutenberg's movable-type printing press.

4.2. The emergence of the Republic of Letters

The appearance of the printer gave rise to accessible and available knowledge for communities. As Eisenstein proved in *The Printing Press As An Agent of Change*, when the works of Copernicus, Vesalius or Bacon were printed with high precision in both text and technical drawings, they created the Republic of Letters [13].

The Republic of Letters was a transnational intellectual community in early modern Europe that connected scholars, scientists, and philosophers through correspondence and shared ideas. This network allowed scientists to verify one another's findings, accumulate knowledge and prevent the loss of information. Furthermore, printing books in native language instead of using only Latin weakened the Church's monopoly, promoting critical thinking and the formation of social opinion.

The Republic of Letters influenced modern life by creating one of the first international networks for sharing knowledge. It encouraged intellectual collaboration, critical thinking, and the free exchange of ideas, which later became essential foundations of modern science, education, and democratic societies.

5. Conclusion

In conclusion, through the rise of humanism, the astronomical discoveries of Copernicus and Galileo, the anatomical studies of Vesalius and Leonardo da Vinci, also the dissemination of knowledge enabled by Gutenberg's printing press, the Renaissance challenged long-established assumptions about the world. These developments not only transformed the production of knowledge but also laid the foundations for the Scientific Revolution and the emergence of modern scientific thought.

Beyond its historical achievements, the Renaissance continues to shape modern science and society. By promoting rational observation and critical thinking, it established principles that remain at the core of modern scientific research. At the same time, its humanistic ideals encouraged individual freedom, intellectual curiosity, and confidence in human potential, values that continue to influence education, democratic institutions and concepts of human rights. Therefore, the Renaissance not only paved the way for the Enlightenment and the Industrial Revolutions that followed but also directly shaped the fundamental values and institutions that define the modern world today.

REFERENCE

- [1] Wikipedia. (2018, November 26). *Great Pyramid of Giza*. Wikipedia; Wikimedia Foundation. https://en.wikipedia.org/wiki/Great_Pyramid_of_Giza
- [2] AncientEpoch. (2025, May 27). *Great Pyramid: Eight Side-Sided Engineering Brilliance - Ancient History X*. Ancient History X. <https://ancienthistoryx.com/great-pyramid-eight-side-sided-engineering-brilliance/>
- [3] Lehner, M. (1997). *The Complete Pyramids*. Thames & Hudson.
- [4] Hawass, Z. (2006). *Mountains of Pharaoh: The Untold Story of the Pyramid Builders*. Doubleday.
- [5] Murray, M. A. (2004). "Provisions for the pyramid builders: new evidence from the ancient site of Giza." *Archaeology International*, 8, 37-40.
- [6] Kemp, B. J. (2006). *Ancient Egypt: Anatomy of a Civilization (2nd ed.)*. Routledge.
- [7] Aquinas, T. (2012). *Summa Theologiae*. The Aquinas Institute For The Study Of Sacred Doctrine. (Original work published 1274)
- [8] Nauert, C. G. (2006). *Humanism and the culture of Renaissance Europe (2nd ed.)*. Cambridge University Press.

- [9] Pico della Mirandola, G. (1956). *Oration on the dignity of man* (A. R. Caponigri, Trans.). Regnery Publishing. (Original work published 1486).
- [10] Kuhn, T. S. (1957). *The Copernican revolution: Planetary astronomy in the development of Western thought*. Harvard University Press.
- [11] Vesalius, A. (2003). *On the fabric of the human body* (W. F. Richardson & J. B. Carman, Trans.). San Francisco Medical Press. (Original work published 1543).
- [12] Clayton, M., & Philo, R. (2012). *Leonardo da Vinci: Anatomist*. Royal Collection Trust.
- [13] Eisenstein, E. L. (1980). *The printing press as an agent of change: Communications and cultural transformations in early modern Europe*. Cambridge University Press.
- [14] Copernicus, N. (1992). *On the revolutions of the heavenly spheres* (E. Rosen, Trans.). Johns Hopkins University Press. (Original work published 1543).
- [15] Galilei, G. (1967). *Dialogue concerning the two chief world systems-Ptolemaic and Copernican* (S. Drake, Trans.; 2nd ed.). University of California Press. (Original work published 1632).

